STRICT INEQUALITIES FOR PRODUCTS OF FILTERS

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Abstract. We answer affirmatively the two conjectures in Open Problem Garden problem 37339 concerning the second, oblique, and classical reloidal products of filters. The cofinite filter Φ on $\mathbb{N}$ satisfies $\Phi \times_F^{\mathrm{RLD}} \Phi = \bot \sqsubset \Phi \ltimes \Phi \sqsubset \Phi \times^{\mathrm{RLD}} \Phi$. The same strict chain holds for the proposed one-sided neighbourhood filter Δ_+ on $\mathbb{R}$. In both examples, two disjoint relations lying above the funcoidal product force the second product to be the least reloid, whereas a column characterisation makes the oblique product proper. The complement of the diagonal separates the oblique and classical products. The accompanying Lean development formalises both examples, including the real-number case.

1. The problem and conventions

Porton's problem [1] asks whether there are filter objects $\mathcal{F}, \mathcal{G}$ such that

$$\mathcal{F} \times_F^{\mathrm{RLD}} \mathcal{G} \sqsubset \mathcal{F} \ltimes \mathcal{G} \sqsubset \mathcal{F} \times^{\mathrm{RLD}} \mathcal{G}, \tag{1}$$

and asks specifically about $\mathcal{F} = \mathcal{G} = \Delta \sqcap \uparrow^{\mathbb{R}} (0, +\infty)$. It also asks the weaker question of strictness of the first inequality alone. We use the definitions in [2], with the existential quantifier in the oblique product as corrected by its author in 2016 [3].

A filter on a set U is a nonempty, upward-closed family of subsets of U closed under finite intersections. Improper filters are allowed; a filter is *proper* when it does not contain $\varnothing$. Following the source, the order is

$$\mathcal{F} \sqsubseteq \mathcal{G} \quad \Longleftrightarrow \quad \mathcal{F} \supseteq \mathcal{G},$$

and $\sqsubset$ denotes its strict part. The least filter $\bot$ is $\mathcal{P}(U)$; the principal filter $\uparrow A$ is $\{S \subseteq U : A \subseteq S\}$.

For filters $\mathcal{F}, \mathcal{G}$ on the same set, write $\mathcal{F} \# \mathcal{G}$ when their meet is proper. Equivalently,

$$\mathcal{F} \# \mathcal{G} \quad \Longleftrightarrow \quad A \cap B \neq \varnothing \quad (A \in \mathcal{F},\ B \in \mathcal{G}). \tag{2}$$

Indeed, their meet is the filter generated by $\mathcal{F} \cup \mathcal{G}$, and every finite intersection of members of that union reduces to an intersection of one member of each filter. A set $X \subseteq U$ is *$\mathcal{F}$-positive* if $\uparrow X \# \mathcal{F}$, that is, if it meets every member of $\mathcal{F}$. We denote the family of such sets by $\mathrm{Pos}(\mathcal{F})$.

A reloid from U to V is represented by a filter on $U \times V$. We write $\mathrm{up}(P)$ for its family of relations. Reloid order is therefore $P \sqsubseteq Q$ if and only if $\mathrm{up}(P) \supseteq \mathrm{up}(Q)$. For a family $J \subseteq \mathcal{P}(U \times V)$, put

$$\mathrm{fil}(J) = \{S \subseteq U \times V : \exists n \geq 0,\ R_1, \ldots, R_n \in J,\ R_1 \cap \cdots \cap R_n \subseteq S\}, \tag{3}$$

with empty intersection $U \times V$. This is the family of the reloid meet of the principal reloids generated by members of J.

For a relation $R \subseteq U \times V$, use the notation

$$R[X] = \{y \in V : \exists x \in X,\ (x, y) \in R\}, \qquad C_y(R) = \{x \in U : (x, y) \in R\}.$$

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2. The generating families

The funcoidal product has the graph

$$\mathcal{X}[\mathcal{F} \times^{\mathrm{FCD}} \mathcal{G}]\mathcal{Y} \quad \iff \quad \mathcal{X} \# \mathcal{F} \text{ and } \mathcal{Y} \# \mathcal{G}.$$

For a relation R, the image of $\mathcal{X}$ under its principal funcoid is the filter generated by $\{R[A] : A \in \mathcal{X}\}$, and

$$\mathcal{X}[\uparrow^{\mathrm{FCD}} R]\mathcal{Y} \quad \iff \quad \mathcal{Y} \# \langle \uparrow^{\mathrm{FCD}} R \rangle \mathcal{X}.$$

Funcoids are ordered by inclusion of these graphs. Define

$$\begin{aligned} I_2(\mathcal{F}, \mathcal{G}) &= \{R : \uparrow^{\mathrm{FCD}} R \sqsupseteq \mathcal{F} \times^{\mathrm{FCD}} \mathcal{G}\}, \\ I_{\ltimes}(\mathcal{F}, \mathcal{G}) &= \{R : \exists B_0 \in \mathcal{G},\ \uparrow^{\mathrm{FCD}} R \sqsupseteq \mathcal{F} \times^{\mathrm{FCD}} \uparrow B_0\}, \\ I_{\square}(\mathcal{F}, \mathcal{G}) &= \{A \times B : A \in \mathcal{F},\ B \in \mathcal{G}\}. \end{aligned}$$

The definitions of the three products give

$$\begin{aligned} \mathrm{up}(\mathcal{F} \times_F^{\mathrm{RLD}} \mathcal{G}) &= \mathrm{fil}(I_2(\mathcal{F}, \mathcal{G})), \\ \mathrm{up}(\mathcal{F} \ltimes \mathcal{G}) &= \mathrm{fil}(I_{\ltimes}(\mathcal{F}, \mathcal{G})), \\ \mathrm{up}(\mathcal{F} \times^{\mathrm{RLD}} \mathcal{G}) &= \mathrm{fil}(I_{\square}(\mathcal{F}, \mathcal{G})). \end{aligned} \tag{4}$$

In the first line, the defining operation $(\mathrm{RLD})_{\mathrm{out}} f = \bigsqcap^{\mathrm{RLD}} \mathrm{up}(f)$ ranges over the *principal relations above* f.

Lemma 2.1. *For a filter $\mathcal{F}$ on U and a set $C \subseteq U$,*

$$C \in \mathcal{F} \quad \iff \quad C \cap X \neq \varnothing \text{ for every } X \in \mathrm{Pos}(\mathcal{F}).$$

Proof. The forward implication is the definition of positivity. If $C \notin \mathcal{F}$, no member of $\mathcal{F}$ is contained in C. Thus $U \setminus C$ is $\mathcal{F}$-positive and disjoint from C, proving the reverse implication. □

Lemma 2.2. *For filters $\mathcal{F}$ on U, $\mathcal{G}$ on V, and a relation $R \subseteq U \times V$:*

$$R \in I_2(\mathcal{F}, \mathcal{G}) \iff R[X] \cap Y \neq \varnothing \quad (X \in \mathrm{Pos}(\mathcal{F}),\ Y \in \mathrm{Pos}(\mathcal{G})); \tag{5}$$

$$R \in I_{\ltimes}(\mathcal{F}, \mathcal{G}) \iff \exists B_0 \in \mathcal{G}\ \forall y \in B_0 :\ C_y(R) \in \mathcal{F}. \tag{6}$$

Proof. For (5), necessity follows by testing the graph order on principal filters $\uparrow X, \uparrow Y$. Conversely, if $\mathcal{X} \# \mathcal{F}$ and $\mathcal{Y} \# \mathcal{G}$, every $A \in \mathcal{X}$ is $\mathcal{F}$-positive and every $B \in \mathcal{Y}$ is $\mathcal{G}$-positive. Hence $R[A] \cap B$ is nonempty for all such members. The images $R[A]$ form a filter base: $R[A_1 \cap A_2] \subseteq R[A_1] \cap R[A_2]$. Therefore $\mathcal{Y}$ meshes with the image filter, giving the required graph inclusion.

For a fixed B_0, the same graph condition with second factor $\uparrow B_0$ is equivalent to

$$B_0 \subseteq R[X] \quad \text{for every } X \in \mathrm{Pos}(\mathcal{F}). \tag{7}$$

Necessity is obtained by testing singleton output filters $\uparrow \{y\}$ for $y \in B_0$. For sufficiency, any set positive for $\uparrow B_0$ meets B_0, so it meets $R[X]$. Finally, (7) says that each column $C_y(R)$, $y \in B_0$, meets every $\mathcal{F}$-positive set. Lemma 2.1 identifies this with membership in $\mathcal{F}$. □

Proposition 2.3. *The family $I_{\ltimes}(\mathcal{F}, \mathcal{G})$ is a filter. If both inputs are proper, it is proper. In particular*

$$\mathrm{up}(\mathcal{F} \ltimes \mathcal{G}) = I_{\ltimes}(\mathcal{F}, \mathcal{G}).$$

Moreover, for all filters the non-strict chain corresponding to (1) *holds.*

Proof. The full relation has witness $B_0 = V$, and upward closure follows by enlarging columns. For two relations with witnesses B_1, B_2, use $B_1 \cap B_2 \in \mathcal{G}$. On that witness set, the columns of their intersection belong to $\mathcal{F}$, since $C_y(R_1 \cap R_2) = C_y(R_1) \cap C_y(R_2)$. This proves intersection closure. For proper inputs the witness set is nonempty, and every one of its columns is nonempty; consequently no member of the family is empty.

Every rectangle in $I_\square$ belongs to $I_\ltimes$, with its right side as witness. If $R \in I_\ltimes$ has witness B_0, then every $\mathcal{G}$-positive Y meets B_0, and every $\mathcal{F}$-positive X satisfies $B_0 \subseteq R[X]$. Thus $I_\ltimes \subseteq I_2$. Taking generated families and reversing inclusion proves the chain. □

3. The cofinite filter

Let $\Phi = \{A \subseteq \mathbb{N} : \mathbb{N} \setminus A \text{ is finite}\}$. It is proper and non-principal, and $\mathrm{Pos}(\Phi)$ consists exactly of the infinite subsets of $\mathbb{N}$. Lemmas 2.1 and 2.2 give

$$R \in I_2(\Phi, \Phi) \iff R[X] \text{ is cofinite for every infinite } X,$$
$$R \in I_\ltimes(\Phi, \Phi) \iff \text{cofinitely many columns of } R \text{ are cofinite.}$$

Theorem 3.1. *The proper, non-principal input filter Φ satisfies*

$$\Phi \times_F^{\mathrm{RLD}} \Phi = \bot \sqsubset \Phi \ltimes \Phi \sqsubset \Phi \times^{\mathrm{RLD}} \Phi.$$

Proof. Put $L = \{(x, y) \in \mathbb{N}^2 : y < x\}$ and $H = \{(x, y) \in \mathbb{N}^2 : x < y\}$. For infinite X, unboundedness gives $L[X] = \mathbb{N}$, whereas $H[X] = \{y : \min X < y\}$ is cofinite. Thus $L, H \in I_2(\Phi, \Phi)$. Since $L \cap H = \varnothing$, the second product's generated family contains the empty relation and is the improper filter. Proposition 2.3 makes the oblique product proper, so the first inequality is strict.

Let $D = \{(x, y) \in \mathbb{N}^2 : x \neq y\}$. Every column of D is cofinite; hence $D \in \mathrm{up}(\Phi \ltimes \Phi)$. On the other hand, every cofinite rectangle $A \times B$ contains (n, n) for some $n \in A \cap B$. Thus no such rectangle is contained in D. Finite intersections of cofinite rectangles are again cofinite rectangles, so $D \notin \mathrm{up}(\Phi \times^{\mathrm{RLD}} \Phi)$. With the non-strict chain, this proves strictness of the second inequality. □

Corollary 3.2. *Both existential conjectures in Open Problem Garden problem 37339 are true.*

4. The proposed real-number example

On $\mathbb{R}$, let

$$\Delta = \{S \subseteq \mathbb{R} : (-\varepsilon, \varepsilon) \subseteq S \text{ for some } \varepsilon > 0\}.$$

This is the neighbourhood filter of 0, equivalently the filter meet of the principal symmetric intervals in the problem's source. Its meet with the principal positive half-line is

$$\Delta_+ = \Delta \sqcap \uparrow^{\mathbb{R}} (0, +\infty) = \{S \subseteq \mathbb{R} : \exists \varepsilon > 0,\ (0, \varepsilon) \subseteq S\}. \tag{8}$$

The family in (8) is proper because every positive interval is nonempty. It is non-principal because the intersection of all those intervals is empty. Its positive sets satisfy

$$X \in \mathrm{Pos}(\Delta_+) \iff X \cap (0, \varepsilon) \neq \varnothing \quad \text{for every } \varepsilon > 0. \tag{9}$$

Theorem 4.1. *For the proper, non-principal real-number filter in* (8)*,*

$$\Delta_+ \times_F^{\mathrm{RLD}} \Delta_+ = \bot \sqsubset \Delta_+ \ltimes \Delta_+ \sqsubset \Delta_+ \times^{\mathrm{RLD}} \Delta_+.$$

Consequently the particular pair proposed in the problem also has an affirmative answer.

Proof. Use the disjoint relations $L = \{(x, y) \in \mathbb{R}^2 : y < x\}$ and $H = \{(x, y) \in \mathbb{R}^2 : x < y\}$. For a Δ_+-positive set X, choose $x \in X \cap (0, 1)$. The image $L[X]$ contains $(0, x)$, a member of Δ_+. For every $y > 0$, choose $x \in X \cap (0, y)$; then $y \in H[X]$. Thus $H[X]$ contains the positive half-line, also a member of Δ_+. Both images meet every Δ_+-positive output

set, so Lemma 2.2 gives $L, H \in I_2(\Delta_+, \Delta_+)$. Their disjointness again forces the second product to be $\perp$, and the oblique product is proper by Proposition 2.3.

For $D = \{(x, y) \in \mathbb{R}^2 : x \neq y\}$, every column $\mathbb{R} \setminus \{y\}$ contains a positive interval: use $(0, y)$ when $y > 0$ and $(0, 1)$ otherwise. Hence all columns belong to Δ_+, and D belongs to the oblique family with witness $\mathbb{R}$. Any two members A, B of Δ_+ intersect, so their rectangle contains a diagonal point and cannot be contained in D. As finite intersections of such rectangles remain rectangles of members, D is absent from the classical family. This proves the second strict inequality. □

Remark 4.2. The proof uses only finite intersection closure of filters. The collapse does not assert that a meet of the corresponding funcoids is the principal funcoid of the empty relation: it is the meet in the *reloid* lattice that is computed by intersecting relation generators. Both input filters are proper; the least reloid is permitted as an output of the second product.

5. Formal verification

The accompanying `lean/StrictProducts.lean` uses Lean 4.27.0 and a pinned Mathlib revision [4]. Filters and reloids are represented by Mathlib's `Filter` type, with its reverse-inclusion order, so their families are necessarily nonempty. The development defines the required funcoids as pairs of filter-valued maps satisfying the meshing equivalence, constructs the principal and product funcoids, and defines the three reloid products from their generating families. It proves the reduction lemmas and the chain from those definitions.

The final statements `frechet_solution` and `real_solution` include properness, non-principality, collapse of the second product, and both strict inequalities. The theorem `deltaPlus_eq_inf` identifies the real filter with the exact meet in (8); `delta_eq_iInf` identifies Δ with the source's meet of symmetric intervals. The real example uses Mathlib's `Real` type, not a rational surrogate. A separate `lean/Check.lean` checks the headline statement shapes and their axiom dependencies. The proof does not require the rest of AGT's composition or category theory.

AI disclosure

The writing of the proof was assisted by DeepSeek 4.1 Flash and OpenAI GPT-6.1 sol.

References

[1] V. Porton, *Strict inequalities for products of filters*, Open Problem Garden, problem 37339, posted August 9, 2011.
`https://www.openproblemgarden.org/op/strict_inequalities_for_products_of_filters`.
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[2] V. Porton, *General Topology as Ordered Semigroup Actions*, Volume 1 (also circulated as *Algebraic General Topology*). Source revision `89eeff3457d1a91c8946caffbd9ff6d8d39142dc`. Relevant sections: filters; funcoidal and reloidal products; reloids induced by a funcoid; and "Second product. Oblique product". `https://github.com/vporton/algebraic-general-topology`.

[3] V. Porton, *A typo in my math book*, December 1, 2016.
`https://math.portonvictor.org/2016/12/01/typo-in-my-math-book/`. The corresponding correction is source commit `bd2992f473603e88d4bd32b314787101557f405c`.

[4] The Mathlib community, *Mathlib*, Lean mathematical library, release `v4.27.0`, revision `a3a10db0e9d66acbebf76c5e6a135066525ac900`.
`https://github.com/leanprover-community/mathlib4`.